Dever de Casa – 05

1) $T(n) = aT(n/b) + f(n)$

→ Merge sort → $a = 2$

→ $b = 2$

→ $f(n) = n$

→ $T(n) = 2T(n/2) + n$

→ $n^{\log_b a} = n^{\log_2 2} = \boxed{n^1 = f(n)}$

↓

$\boxed{T(n) = \Theta(n \log n)}$

2) $T(n) = aT(n/b) + f(n)$

→ Em uma multiplicação tradicional de matrizes $(N \times N)$, podemos quebrar cada uma em 4 submatrizes de tamanho $N/2$.

→ $a = 8$

$b = 2$

$f(n) = n^2$

→ $T(n) = 8T(n/2) + n^2$

→ $n^{\log_2 8} = n^3 \longrightarrow \boxed{n^3 > f(n)}$

↓

$\boxed{T(n) = \Theta(n^3)}$

3) a) $T(n) = 2T(n/4) + \sqrt{n}$

→ $a = 2;\ b = 4 \longrightarrow n^{\log_4 2} = \boxed{n^{1/2} = f(n)} \longrightarrow \boxed{T(n) = \Theta(\sqrt{n} \log n)}$

b) $T(n) = 2T(n/4) + n$

→ $a = 2;\ b = 4 \longrightarrow n^{\log_4 2} = \boxed{n^{1/2} < f(n)} \longrightarrow \boxed{T(n) = \Theta(n)}$

c) $T(n) = 16T(n/4) + n^2 \longrightarrow n^{\log_4 16} = \boxed{n^2 = f(n)} \longrightarrow \boxed{T(n) = \Theta(n^2 \log n)}$